Class 12 · Introduction to Symbolization

PHIL 270 · Symbolic Logic · Practice sheet. Circle the best answer. The answer key is on the last page.

Name: Date:

Definitions

Symbolization key. A list that says what we are talking about (the UD), gives the English reading of each predicate, and says which member of the UD each constant names.

UD: the members of a book club Lxy: x likes y a: Ana With this key, Lab says "Ana likes Ben."

Universe of discourse (UD). The set of things we are talking about. A UD is never empty. We can list its members in curly brackets, or describe it in words.

UD: {Ana, Ben, Cara} UD: the positive integers

Variables in a key only mark places. Writing "Lxy: x likes y" does not mean you must use x and y. Any variables, or constants, can fill the two places.

$(\forall x)Lxa \quad = (\forall z)Lza$ Both say "Everyone likes Ana."

Small UD: no quantifiers needed. When the UD has only a few members, "everyone" is a conjunction and "no one" is a negated disjunction. This stops working when the UD is large or infinite, which is why we need quantifiers.

$(Fa \,\&\, Fb) \,\&\, Fc \quad \sim[(Fa \vee Fb) \vee Fc]$ UD: {Ana, Ben, Cara}. Everyone is friendly. No one is friendly.

Paraphrase first. Rewrite the English with "Each x is such that…" for ∀ and "There is an x such that…" for ∃. Then symbolize the paraphrase piece by piece.

$(\forall x)Lxa$ Everyone likes Ana → Each x is such that x likes Ana.

Every P is Q: ∀ with ⊃. To say every member of a kind has a property, use a universal quantifier with a conditional. It says nothing about things that are not P.

$(\forall x)(Dx \supset Bx)$ Every dog barks.

Some P is Q: ∃ with &. To say at least one member of a kind has a property, use an existential quantifier with a conjunction.

$(\exists x)(Dx \,\&\, Bx)$ Some dog barks.

The wrong pairings. ∀ with & says everything is both P and Q, which is far too strong. ∃ with ⊃ is true as soon as one thing is not P, which is far too weak. When you see these shapes, check again.

$(\forall x)(Dx \,\&\, Bx) \quad (\exists x)(Dx \supset Bx)$

"Any" and "anyone". Usually universal. But when "anyone" sits in the "if" part of a sentence whose "then" part is about someone else, the whole sentence is a conditional with ∃ in the antecedent.

$(\forall x)(Lxa \supset Lxb) \quad (\exists x)Lxa \supset Lba$ Anyone who likes Ana likes Ben. If anyone likes Ana, Ben does.

Two quantifiers. Paraphrase from the outside in. Each quantifier gets its own variable, and each kind gets its own ⊃ or &.

$(\forall x)[Dx \supset (\exists y)(Cy \,\&\, Pxy)]$ Every dog plays with some cat.

A, E, I, and O sentences. Four classic quantity claims. A: All P are Q. E: No P are Q. I: Some P are Q. O: Some P are not Q.

$(\forall x)(Px \supset Qx) \quad (\forall x)(Px \supset \sim Qx) \quad (\exists x)(Px \,\&\, Qx)$

$(\exists x)(Px \,\&\, \sim Qx)$ In order: A, E, I, O.

Contradictories. On the square of opposition, A and O are contradictories, and so are E and I: each one is equivalent to the negation of the other.

$(\forall x)(Px \supset Qx) \quad = \sim(\exists x)(Px \,\&\, \sim Qx)$ All P are Q = It is not the case that some P is not Q.

Part 1: Symbolization keys

Using a symbolization key

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

1. Ana likes Ben, and Ben likes Cara.

A $Lab \vee Lbc$ **B** $Lab \,\&\, Lcb$

C $Lab \,\&\, Lbc$ **D** $Lba \,\&\, Lcb$

2. If Cara respects Ana, then Ana likes Cara.

A $Lac \supset Rca$ **B** $Rca \supset Lac$

C $Rac \supset Lca$ **D** $Rca \,\&\, Lac$

3. Which English sentence does $Lba \equiv (Lab \,\&\, \sim Lac)$ say?

A Ben likes Ana, Ana likes Ben, and Ana does not like Cara.

B Ben likes Ana if Ana likes Ben and does not like Cara.

C Ben likes Ana if and only if Ana likes Ben and does not like Cara.

D Ana likes Ben if and only if Ben likes Ana and does not like Cara.

UD: {Ana, Ben, Cara} · **Fx**: x is friendly · **a**: Ana · **b**: Ben · **c**: Cara

4. Everyone is friendly. (Symbolize it without quantifiers.)

A $(Fa \vee Fb) \vee Fc$ **B** $Fa \,\&\, Fb$

C $\sim[(Fa \vee Fb) \vee Fc]$ **D** $(Fa \,\&\, Fb) \,\&\, Fc$

5. No one is friendly. (Symbolize it without quantifiers.)

A $\sim[(Fa \,\&\, Fb) \,\&\, Fc]$ **B** $\sim[(Fa \vee Fb) \vee Fc]$

C $\sim Fa \,\&\, \sim Fb$ **D** $(\sim Fa \vee \sim Fb) \vee \sim Fc$

6. Why do we need quantifiers at all, if "everyone" can be a conjunction?

A Constants can only name people

B Conjunctions are not allowed inside predicate logic

C A UD is allowed to be empty

D When the UD is large or infinite, like the positive integers, no conjunction can list every member

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

7. Neither Ana nor Ben likes Cara.

A $\sim Lca \,\&\, \sim Lcb$ **B** $\sim (Lac \,\&\, Lbc)$

C $\sim Lac \,\&\, \sim Lbc$ **D** $\sim Lac \vee \sim Lbc$

8. Ana likes Ben only if Ben respects Ana.

A $Lab \supset Rab$ **B** $Lab \equiv Rba$

C $Rba \supset Lab$ **D** $Lab \supset Rba$

Read it in English

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

11. Which English sentence does $(\forall x)Lax$ say?

A Everyone likes Ana. **B** Ana likes someone.

C Ana likes herself. **D** Ana likes everyone.

12. Which English sentence does $(\exists x)(Lxb \,\&\, \sim Rxb)$ say?

A Someone likes Ben, and someone does not respect him.

B Someone likes Ben but does not respect him.

C Everyone who likes Ben does not respect him.

D No one who likes Ben respects him.

13. Which English sentence does $\sim (\exists x)Lxc$ say?

A Cara likes no one. **B** Someone does not like Cara.

C Not everyone likes Cara. **D** No one likes Cara.

9. Cara likes herself but does not respect herself.

A $Lcc \vee \sim Rcc$ **B** $Lc \,\&\, \sim Rc$

C $Lcc \,\&\, \sim Rca$ **D** $Lcc \,\&\, \sim Rcc$

10. Ben likes Ana unless Ana likes Cara.

A $Lba \vee Lac$ **B** $Lab \vee Lca$

C $Lac \supset Lba$ **D** $Lba \,\&\, Lac$

14. Which English sentence does $(\exists x)Lxa \supset Lba$ say?

A Someone likes Ana, and Ben likes Ana.

B If everyone likes Ana, Ben does.

C If Ben likes Ana, someone does.

D If anyone likes Ana, Ben does.

15. Which English sentence does $(\forall x)(Rxc \supset Lxc)$ say?

A Everyone who likes Cara respects her.

B Cara likes everyone who respects her.

C Everyone who respects Cara likes her.

D Everyone respects Cara and likes her.

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Bx**: x barks · **Sx**: x is asleep

16. Which English sentence does $\sim (\exists x)(Cx \,\&\, Bx)$ say?

A Only cats bark. **B** Some cat does not bark.

C No cat barks. **D** Not every cat barks.

Part 2: One quantifier

One quantifier

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

17. Everyone likes Ana.

A Lxa **B** $(\exists x)Lxa$

C $(\forall x)Lxa$ **D** $(\forall x)Lax$

18. Someone likes Ben, and someone does not like Ben.

A $(\exists x)(Lxb \,\&\, \sim Lxb)$ **B** $(\forall x)Lxb \,\&\, (\exists y) \sim Lyb$

C $(\exists x)Lxb \,\&\, (\exists y) \sim Lyb$ **D** $(\exists x)Lxb \,\&\, \sim (\exists x)Lxb$

19. Everyone who likes Ana also respects her.

A $(\forall x)(Lxa \,\&\, Rxa)$ **B** $(\forall x)(Lxa \supset Rxa)$

C $(\exists x)(Lxa \,\&\, Rxa)$ **D** $(\forall x)(Rxa \supset Lxa)$

20. Someone likes Cara and respects her.

A $(\exists x)(Lxc \supset Rxc)$ **B** $(\exists x)(Lxc \,\&\, Rxc)$

C $(\forall x)(Lxc \supset Rxc)$ **D** $(\exists x)Lxc \,\&\, (\exists x)Rxc$

21. If anyone respects Ben, then Cara does.

A $(\forall x)(Rxb \,\&\, Rcb)$ **B** $(\exists x)Rxb \supset Rcb$

C $(\exists x)(Rxb \supset Rcb)$ **D** $(\forall x)Rxb \supset Rcb$

22. Which sentence is equivalent to $(\exists x)Rxb \supset Rcb$?

A $(\forall x)Rxb \supset Rcb$ **B** $(\forall x)(Rxb \supset Rcb)$

C $(\forall x)(Rcb \supset Rxb)$ **D** $(\exists x)(Rxb \supset Rcb)$

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Bx**: x barks · **Sx**: x is asleep

23. The dogs are asleep. (We are talking about the animals at one shelter.)

A $(\forall x)(Dx \supset Sx)$ **B** $(\forall x)(Dx \,\&\, Sx)$

C $(\exists x)(Dx \,\&\, Sx)$ **D** Sd, where d: the dogs

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

24. Ana likes everyone who likes her.

A $(\exists x)(Lxa \,\&\, Lax)$ **B** $(\forall x)(Lax \supset Lxa)$

C $(\forall x)(Lxa \,\&\, Lax)$ **D** $(\forall x)(Lxa \supset Lax)$

25. Not everyone likes Ben.

A $\sim (\exists x)Lxb$ **B** $\sim (\forall x)Lxb$

C $(\forall x) \sim Lxb$ **D** $\sim (\forall x)Lbx$

26. Ben likes no one.

A $\sim (\forall x)Lbx$ **B** $\sim (\exists x)Lxb$

C $(\forall x) \sim Lbx$ **D** $(\forall x) \sim Lxb$

27. No one who respects Ana likes Cara.

A $(\forall x)(Rxa \supset \sim Lxc)$ **B** $(\exists x)(Rxa \,\&\, \sim Lxc)$

C $\sim (\forall x)(Rxa \supset Lxc)$ **D** $(\forall x)(\sim Rxa \supset Lxc)$

28. Everyone likes Ben, or everyone likes Cara.

A $(\exists x)(Lxb \vee Lxc)$ **B** $(\forall x)(Lxb \,\&\, Lxc)$
C $(\forall x)Lxb \vee (\forall y)Lyc$ **D** $(\forall x)(Lxb \vee Lxc)$

29. Which sentence is true no matter who likes whom?

A $Lba \supset (\forall x)Lxa$ **B** $(\forall x)(Lxa \supset Lba)$
C $(\exists x)Lxa \supset Lba$ **D** $(\forall x)Lxa \supset Lba$

Part 3: All, no, and some

All, no, and some

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Bx**: x barks · **Sx**: x is asleep

30. No cat barks.

A $(\exists x)(Cx \,\&\, \sim Bx)$ **B** $(\forall x)(\sim Cx \supset Bx)$
C $(\forall x)(Cx \supset \sim Bx)$ **D** $\sim(\forall x)(Cx \supset Bx)$

31. Some dogs are not asleep.

A $(\exists x)(Dx \supset \sim Sx)$ **B** $(\exists x)(Dx \,\&\, \sim Sx)$
C $\sim(\exists x)(Dx \,\&\, Sx)$ **D** $(\forall x)(Dx \supset \sim Sx)$

32. Which kind of sentence is “Some dogs are asleep”?

A A: All P are Q **B** E: No P are Q
C I: Some P are Q **D** O: Some P are not Q

33. Neither the dogs nor the cats are asleep.

A $(\forall x)[(Dx \vee Cx) \supset \sim Sx]$ **B** $(\forall x)[(Dx \,\&\, Cx) \supset \sim Sx]$
C $(\exists x)[(Dx \vee Cx) \,\&\, \sim Sx]$ **D** $(\forall x)(Dx \supset \sim Sx) \vee (\forall y)(Cy \supset \sim Sy)$

34. Which sentence is the contradictory of “All dogs bark,” $(\forall x)(Dx \supset Bx)$?

A $(\forall x)(Dx \supset \sim Bx)$ **B** $(\exists x)(Dx \,\&\, \sim Bx)$
C $(\exists x)(Dx \,\&\, Bx)$ **D** $(\forall x)(\sim Dx \supset \sim Bx)$

35. Which sentence is equivalent to “No cat is asleep,” $(\forall x)(Cx \supset \sim Sx)$?

A $\sim(\forall x)(Cx \supset Sx)$ **B** $\sim(\exists x)(Cx \,\&\, Sx)$
C $(\exists x)(Cx \,\&\, \sim Sx)$ **D** $\sim(\exists x)(Cx \,\&\, \sim Sx)$

36. Only dogs bark.

A $(\forall x)(Bx \supset Dx)$ **B** $(\forall x)(Dx \supset Bx)$
C $(\forall x)(Dx \,\&\, Bx)$ **D** $(\exists x)(Bx \,\&\, Dx)$

37. Not all dogs bark.

A $\sim(\exists x)(Dx \,\&\, Bx)$ **B** $(\forall x)(Dx \supset \sim Bx)$
C $\sim(\forall x)(Dx \supset Bx)$ **D** $(\forall x) \sim (Dx \supset Bx)$

38. All dogs and cats are asleep.

A $(\forall x)[(Dx \vee Cx) \supset Sx]$ **B** $(\forall x)[(Dx \,\&\, Cx) \supset Sx]$
C $(\forall x)[(Dx \vee Cx) \,\&\, Sx]$ **D** $(\exists x)[(Dx \vee Cx) \,\&\, Sx]$

39. Some dog that barks is asleep.

A $(\exists x)(Dx \,\&\, Bx) \,\&\, (\exists y)Sy$ **B** $(\forall x)[(Dx \,\&\, Bx) \supset Sx]$
C $(\exists x)[(Dx \,\&\, Bx) \supset Sx]$ **D** $(\exists x)[(Dx \,\&\, Bx) \,\&\, Sx]$

40. Which kind of sentence is “No cats bark”?

A A: All P are Q **B** E: No P are Q
C I: Some P are Q **D** O: Some P are not Q

41. Which sentence is the contradictory of “Some cats are asleep,” $(\exists x)(Cx \,\&\, Sx)$?

A $(\exists x)(Cx \,\&\, \sim Sx)$ **B** $(\forall x)(Cx \supset \sim Sx)$
C $(\exists x)(\sim Cx \,\&\, Sx)$ **D** $(\forall x)(Cx \supset Sx)$

42. Which sentence is equivalent to “Not every dog is asleep,” $\sim(\forall x)(Dx \supset Sx)$?

A $(\forall x)(Dx \supset \sim Sx)$ **B** $\sim(\exists x)(Dx \,\&\, \sim Sx)$
C $(\exists x)(Dx \,\&\, Sx)$ **D** $(\exists x)(Dx \,\&\, \sim Sx)$

Part 4: Two quantifiers and relations

Quantifier order

UD: people · Write the formula. With mixed quantifiers, order matters.

43. Everyone likes someone. (**Lxy**: x likes y)

44. There is someone whom everyone admires. (**Axy**: x admires y)

45. No one knows everyone. (**Kxy**: x knows y)

46. Everyone is trusted by someone. (**Txy**: x trusts y)

47. Someone likes no one. (**Lxy**: x likes y)

48. No one is admired by everyone. (**Axy**: x admires y)

Two quantifiers

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Pxy**: x plays with y · **Hxy**: x is heavier than y

49. Every animal plays with every animal.

A $(\forall x)Pxx$ **B** $(\exists x)(\forall y)Pxy$
C $(\forall x)(\forall y)Pxy$ **D** $(\forall x)(\exists y)Pxy$

50. There is an animal that plays with every animal.

A $(\exists x)(\forall y)Pyx$ **B** $(\forall x)(\exists y)Pxy$
C $(\forall y)(\exists x)Pxy$ **D** $(\exists x)(\forall y)Pxy$

51. No animal is heavier than itself.

A $\sim(\exists x)(\exists y)Hxy$ **B** $(\exists x) \sim Hxx$
C $(\forall x)(\forall y) \sim Hxy$ **D** $\sim(\exists x)Hxx$

52. If one animal is heavier than another, the second is not heavier than the first.

A $(\forall x)(\forall y)(Hxy \supset Hyx)$ **B** $(\exists x)(\exists y)(Hxy \,\&\, \sim Hyx)$
C $(\forall x)(\forall y)(Hxy \supset \sim Hyx)$ **D** $(\forall x)(\forall y)(Hxy \supset \sim Hxy)$

53. Every dog plays with every cat.

A $(\forall x)[Dx \supset (\forall y)(Cy \,\&\, Pxy)]$ **B** $(\forall x)[Dx \supset (\exists y)(Cy \,\&\, Pxy)]$
C $(\forall x)(\forall y)[(Dx \,\&\, Cy) \,\&\, Pxy]$ **D** $(\forall x)[Dx \supset (\forall y)(Cy \supset Pxy)]$

54. Every dog plays with at least one cat.

A $(\forall x)[Dx \supset (\exists y)(Cy \,\&\, Pxy)]$ **B** $(\exists y)[Cy \,\&\, (\forall x)(Dx \supset Pxy)]$
C $(\forall x)[Dx \supset (\exists y)(Cy \supset Pxy)]$ **D** $(\forall x)[Dx \,\&\, (\exists y)(Cy \,\&\, Pxy)]$

55. Some dog plays with no cat.

A $(\exists x)[Dx \,\&\, (\exists y)(Cy \,\&\, \sim Pxy)]$ **B** $(\exists x)[Dx \,\&\, \sim (\exists y)(Cy \,\&\, Pxy)]$
C $(\exists x)[Dx \,\&\, (\forall y)(Cy \,\&\, \sim Pxy)]$ **D** $\sim (\exists x)[Dx \,\&\, (\exists y)(Cy \,\&\, Pxy)]$

56. Which English sentence does $(\forall x)[Cx \supset (\forall y)(Dy \supset \sim Pxy)]$ say?

A No cat plays with any dog.
B No dog plays with any cat.
C Not every cat plays with every dog.
D Some cat does not play with some dog.

57. Some dog plays with every cat.

A $(\exists x)[Dx \supset (\forall y)(Cy \supset Pxy)]$ **B** $(\exists x)[Dx \,\&\, (\forall y)(Cy \supset Pxy)]$
C $(\forall y)[Cy \supset (\exists x)(Dx \,\&\, Pxy)]$ **D** $(\exists x)[Dx \,\&\, (\forall y)(Cy \,\&\, Pxy)]$

58. Every cat is heavier than some dog.

A $(\exists y)[Dy \,\&\, (\forall x)(Cx \supset Hxy)]$ **B** $(\forall x)(\forall y)[(Cx \,\&\, Dy) \supset Hxy]$
C $(\forall x)[Cx \supset (\exists y)(Dy \,\&\, Hyx)]$ **D** $(\forall x)[Cx \supset (\exists y)(Dy \,\&\, Hxy)]$

59. No dog plays with itself.

A $\sim (\exists x)(\exists y)(Dx \,\&\, Pxy)$ **B** $\sim (\exists x)(Dx \,\&\, Pxx)$
C $(\forall x)(Dx \supset \sim Pxy)$ **D** $(\exists x)(Dx \,\&\, \sim Pxx)$

60. Every animal that plays with a cat is a dog.

A $(\forall x)[Dx \supset (\exists y)(Cy \,\&\, Pxy)]$ **B** $(\forall x)[(\exists y)(Cy \supset Pxy) \supset Dx]$
C $(\forall x)[(\exists y)(Cy \,\&\, Pxy) \supset Dx]$ **D** $(\exists x)[(\exists y)(Cy \,\&\, Pxy) \,\&\, Dx]$

61. There is a cat that no dog plays with.

A $(\exists x)[Cx \,\&\, \sim (\exists y)(Dy \,\&\, Pyx)]$ **B** $(\exists x)[Cx \,\&\, (\exists y)(Dy \,\&\, \sim Pyx)]$
C $\sim (\exists x)[Cx \,\&\, (\exists y)(Dy \,\&\, Pyx)]$ **D** $(\exists x)[Cx \,\&\, \sim (\exists y)(Dy \,\&\, Pxy)]$

62. Which English sentence does $(\forall x)[Dx \supset (\exists y)(Cy \,\&\, Hyx)]$ say?

A Some cat is heavier than every dog.
B Every dog is heavier than some cat.
C Every cat is heavier than some dog.
D For every dog, some cat is heavier than it.

True in the club?

UD: {Ana, Ben, Cara} · **Lxy**: x likes y · **a**: Ana · **b**: Ben · **c**: Cara · **The only facts: Ana likes Ben, Ben likes himself, Cara likes Ben, and Ben likes Cara. No one else likes anyone.**

63. Is $(\forall x)Lxb$ true or false?

A True **B** False

64. Is $(\forall x)Lbx$ true or false?

A True **B** False

65. Is $(\exists x)(\forall y)Lyx$ true or false?

A True **B** False

66. Is $(\forall x)(\exists y)Lxy$ true or false?

A True **B** False

67. Is $(\forall x)(\exists y)Lyx$ true or false?

A True **B** False

68. Is $(\exists x)Lxa$ true or false?

A True **B** False

69. Is $(\exists x)Lxa \supset Lba$ true or false?

A True **B** False

70. Is $(\forall x)(Lxb \supset Lbx)$ true or false?

A True **B** False

The positive integers

UD: the positive integers · **Ex**: x is even · **Ox**: x is odd · **Px**: x is prime · **Lxy**: x is less than y · **b**: 2

71. There is a smallest positive integer.

A $(\forall x)(\exists y)Lyx$ **B** $(\exists x) \sim (\exists y)Lyx$
C $(\exists x)(\forall y)Lxy$ **D** $\sim (\exists x)(\exists y)Lyx$

72. There is no largest positive integer.

A $\sim (\exists x) \sim (\exists y)Lxy$ **B** $\sim (\exists x)(\exists y)Lxy$
C $(\exists x) \sim (\exists y)Lxy$ **D** $(\exists x)(\forall y)Lyx$

73. 2 is prime, and no prime is less than 2.

A $Pb \,\&\, \sim (\exists x)(Px \,\&\, Lxb)$ **B** $(\exists x)(Px \,\&\, \sim Lxb)$
C $Pb \,\&\, \sim (\exists x)(Px \,\&\, Lbx)$ **D** $Pb \,\&\, (\forall x)(Px \supset Lxb)$

74. Every prime greater than 2 is odd.

A $(\forall x)[(Px \,\&\, Lxb) \supset Ox]$ **B** $(\forall x)[(Px \,\&\, Lbx) \supset Ox]$
C $(\forall x)[Px \supset (Lbx \,\&\, Ox)]$ **D** $(\exists x)[(Px \,\&\, Lbx) \,\&\, Ox]$

75. Every even number is less than some odd number.

A $(\forall x)[Ex \supset (\exists y)(Oy \,\&\, Lxy)]$ **B** $(\exists y)[Oy \,\&\, (\forall x)(Ex \supset Lxy)]$
C $(\forall x)(\forall y)[(Ex \,\&\, Oy) \supset Lxy]$ **D** $(\forall x)[Ex \supset (\exists y)(Oy \,\&\, Lyx)]$

76. No positive integer is less than itself.

A $(\forall x)(\forall y) \sim Lxy$ **B** $\sim (\exists x)Lxx$
C $(\exists x) \sim Lxx$ **D** $\sim (\exists x)(\exists y)Lxy$

77. Some prime is even.

A $(\exists x)(Px \,\&\, Ex)$ **B** $(\exists x)(Px \supset Ex)$
C $(\forall x)(Px \supset Ex)$ **D** $Pb \supset Eb$

78. In the positive integers, is $(\forall x)(\exists y)Lyx$ true or false?

A True **B** False

Test your knowledge

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

79. Ben respects Cara, but Cara does not like Ben.

A Rbc ⊃ ~ Lcb
B Rcb & ~ Lbc
C Rbc & ~ Lbc
D Rbc & ~ Lcb

UD: {Ana, Ben, Cara} · **Fx**: x is friendly · **a**: Ana · **b**: Ben · **c**: Cara

80. Someone is friendly. (Symbolize it without quantifiers.)

A Fa ∨ Fb
B (Fa ∨ Fb) ∨ Fc
C ~ [(Fa ∨ Fb) ∨ Fc]
D (Fa & Fb) & Fc

UD: the members of a book club · **Lxy**: x likes y · **Rxy**: x respects y · **a**: Ana · **b**: Ben · **c**: Cara

81. Everyone respects Cara.

A (∀x)Rcx
B Rcc
C (∀x)Rxc
D (∃x)Rxc

82. If anyone likes Ben, Ana does.

A (∀x)Lxb ⊃ Lab
B (∃x)Lxb ⊃ Lab
C (∃x)(Lxb & Lab)
D (∃x)(Lxb ⊃ Lab)

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Bx**: x barks · **Sx**: x is asleep

83. Some cats are asleep.

A (∀x)(Cx ⊃ Sx)
B (∀x)(Cx & Sx)
C (∃x)(Cx ⊃ Sx)
D (∃x)(Cx & Sx)

84. Which sentence is the contradictory of "No dog barks," (∀x)(Dx ⊃ ~ Bx)?

A (∀x)(Dx ⊃ Bx)
B ~ (∃x)(Dx & Bx)
C (∃x)(Dx & Bx)
D (∃x)(Dx & ~ Bx)

85. UD: people. Lxy: x likes y. Which says "There is someone whom everyone likes"?

A (∃x)(∀y)Lxy
B (∀x)(∃y)Lxy
C (∃y)(∀x)Lxy
D (∀y)(∃x)Lxy

UD: the animals at a shelter · **Dx**: x is a dog · **Cx**: x is a cat · **Pxy**: x plays with y · **Hxy**: x is heavier than y

86. Every cat plays with some dog.

A (∀x)[Cx ⊃ (∃y)(Dy ⊃ Pxy)]
B (∀x)[Cx ⊃ (∀y)(Dy ⊃ Pxy)]
C (∃y)[Dy & (∀x)(Cx ⊃ Pxy)]
D (∀x)[Cx ⊃ (∃y)(Dy & Pxy)]

87. No animal is heavier than every animal.

A ~ (∃x)(∀y)Hxy
B (∃x) ~ (∀y)Hxy
C ~ (∃x)Hxx
D (∀x)(∀y) ~ Hxy

UD: the positive integers · **Ex**: x is even · **Ox**: x is odd · **Px**: x is prime · **Lxy**: x is less than y · **b**: 2

88. Every positive integer is less than some prime.

A (∀x)(∃y)(Py ⊃ Lxy)
B (∀x)(∃y)(Py & Lxy)
C (∃y)[Py & (∀x)Lxy]
D (∀x)[Px ⊃ (∃y)Lxy]

Answer key

Using a symbolization key

1. C Lab & Lbc

2. B Rca ⊃ Lac

3. C Ben likes Ana if and only if Ana likes Ben and does not like Cara.

4. D (Fa & Fb) & Fc

5. B ~ [(Fa ∨ Fb) ∨ Fc]

6. D When the UD is large or infinite, like the positive integers, no conjunction can list every member

7. C ~ Lac & ~ Lbc

8. D Lab ⊃ Rba

9. D Lcc & ~ Rcc

10. A Lba ∨ Lac

Read it in English

11. D Ana likes everyone.

12. B Someone likes Ben but does not respect him.

13. D No one likes Cara.

14. D If anyone likes Ana, Ben does.

15. C Everyone who respects Cara likes her.

16. C No cat barks.

One quantifier

17. C (∀x)Lxa

18. C (∃x)Lxb & (∃y) ~ Lyb

19. B (∀x)(Lxa ⊃ Rxa)

20. B (∃x)(Lxc & Rxc)

21. B (∃x)Rxb ⊃ Rcb

22. B (∀x)(Rxb ⊃ Rcb)

23. A (∀x)(Dx ⊃ Sx)

24. D (∀x)(Lxa ⊃ Lax)

25. B ~ (∀x)Lxb

26. C (∀x) ~ Lbx

27. A (∀x)(Rxa ⊃ ~ Lxc)

28. C (∀x)Lxb ∨ (∀y)Lyc

29. D (∀x)Lxa ⊃ Lba

All, no, and some

30. C (∀x)(Cx ⊃ ~ Bx)

31. B (∃x)(Dx & ~ Sx)

32. C I: Some P are Q

33. A (∀x)[(Dx ∨ Cx) ⊃ ~ Sx]

34. B (∃x)(Dx & ~ Bx)

35. B ~ (∃x)(Cx & Sx)

36. A (∀x)(Bx ⊃ Dx)

37. C ~ (∀x)(Dx ⊃ Bx)

38. A (∀x)[(Dx ∨ Cx) ⊃ Sx]

39. D (∃x)[(Dx & Bx) & Sx]

40. B E: No P are Q

41. B (∀x)(Cx ⊃ ~ Sx)

42. D (∃x)(Dx & ~ Sx)

Quantifier order

43. (∀x)(∃y)Lxy

44. (∃x)(∀y)Ayx

45. ~(∃x)(∀y)Kxy

46. (∀x)(∃y)Tyx

47. (∃x)(∀y)~Lxy

48. ~(∃x)(∀y)Ayx

Two quantifiers

49. C (∀x)(∀y)Pxy

50. D (∃x)(∀y)Pxy

51. D ~ (∃x)Hxx

52. C (∀x)(∀y)(Hxy ⊃ ~ Hyx)

53. D (∀x)[Dx ⊃ (∀y)(Cy ⊃ Pxy)]

54. A (∀x)[Dx ⊃ (∃y)(Cy & Pxy)]

55. B (∃x)[Dx & ~ (∃y)(Cy & Pxy)]

56. A No cat plays with any dog.

57. B (∃x)[Dx & (∀y)(Cy ⊃ Pxy)]

58. D (∀x)[Cx ⊃ (∃y)(Dy & Hxy)]

59. B ~ (∃x)(Dx & Pxx)

60. C (∀x)[(∃y)(Cy & Pxy) ⊃ Dx]

61. A (∃x)[Cx & ~ (∃y)(Dy & Pyx)]

62. D For every dog, some cat is heavier than it.

True in the club?

63. A

64. B

65. A

66. A

67. B

68. B

69. A

70. B

The positive integers

71. B (∃x) ~ (∃y)Lyx

72. A ~ (∃x) ~ (∃y)Lxy

73. A Pb & ~ (∃x)(Px & Lxb)

74. B (∀x)[(Px & Lbx) ⊃ Ox]

75. A (∀x)[Ex ⊃ (∃y)(Oy & Lxy)]

76. B ~ (∃x)Lxx

77. A (∃x)(Px & Ex)

78. B

Test your knowledge

79. D Rbc & ~ Lcb

80. B (Fa ∨ Fb) ∨ Fc

81. C (∀x)Rxc

82. B (∃x)Lxb ⊃ Lab

83. D (∃x)(Cx & Sx)

84. C (∃x)(Dx & Bx)

85. C (∃y)(∀x)Lxy

86. D (∀x)[Cx ⊃ (∃y)(Dy & Pxy)]

87. A ~ (∃x)(∀y)Hxy

88. B (∀x)(∃y)(Py & Lxy)